Retrieval-Augmented Generation: An Overview

Retrieval-augmented generation (RAG) is a pattern for grounding a language model's answers in a specific corpus of documents rather than relying solely on what the model memorized during training. Instead of asking the model to answer a question directly, a RAG system first retrieves the passages most relevant to the question from an external index, then asks the model to answer using only that retrieved context.

The main motivation is factual grounding. A language model's parametric knowledge is frozen at training time, can be wrong, and cannot be traced back to a source. Retrieval lets a system answer questions about documents it has never been trained on, keep its knowledge current by simply updating the index, and cite the exact passage an answer came from, which lets a user verify the answer rather than trust it blindly.

A RAG pipeline has three stages that each affect answer quality independently. Chunking splits source documents into passages small enough to embed meaningfully and to fit in the model's context window, but large enough to preserve the surrounding context a sentence needs to be understood; a common strategy is a fixed token or character budget per chunk with a small overlap between consecutive chunks so a fact near a chunk boundary is not split away from the sentence that explains it.

Retrieval embeds the user's question and the corpus chunks into the same vector space and returns the chunks whose embeddings are closest to the question's embedding, usually by cosine similarity. Retrieval quality is measured independently of generation quality, typically with precision and recall at a cutoff k against a hand-labeled set of question and relevant-chunk pairs, since a wrong or missing retrieval dooms the final answer regardless of how good the generation step is.

Generation (or, in a purely extractive system, answer selection) conditions on the retrieved chunks and the original question to produce a final answer. Even a strong generator cannot recover from a retrieval step that missed the relevant passage, which is why production RAG systems invest heavily in retrieval evaluation and treat it as the primary lever for improving end-to-end answer quality, well before tuning the generation step itself.